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A PDE model of clonal plant competition with nonlinear diffusion

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ABSTRACT

We present a spatial competition model for clonal plant growth that combines two different mechanisms of competition. The first one is described by the standard underlying Kolmogorov Model for two interacting populations. A second competition mechanism, more specific to clonal plant growth, expresses the motility of each species and their capacity to resist to competitor's space intrusion. This model leads to a degenerated nonlinear reaction–diffusion system in which the diffusion coefficient of each species vanishes wherever the other species is beyond a certain biomass value.

Indeed, we show that pattern forming instability cannot occur in general competition systems if the exclusion ability of both species is small even if the diffusions are degenerated. The effect of diffusion degeneracy on patterns formation is carried out. Numerical simulations of this two-level competition model are performed when the reaction terms are given by the competitive Lotka–Volterra equations. We, finally, discuss the potential of such nonlinear reaction–diffusion systems to be a surrogate model for phalanx–guerilla species competition.

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1. Introduction

The dynamic of clonal plant population is a fascinating area of research (van Groenendael and de Kroon, 1990; de Kroon and van Groenendael, 1997). Agent based models with explicit representation of plant individual have been the main modeling tool both for understanding mechanism and predicting spatial–temporal behavior (Oborny et al., 2012). On the other hand PDE based model provides information at the macroscopic level, i.e. does not represent explicitly plants individual, but rather represent a density of plants. PDE models are interesting tools to extract global indicators on competition such as speed of propagation or space and time scale for dynamic. Up to our knowledge there has been no PDE model specifically designed to model clonal plants. There is, however, a vast literature on PDE modeling of the dynamic of population in general.

The role of spatial heterogeneities and dispersal for chemically reacting species or biologically interacting populations, in the linear or nonlinear regime, has been the subject of a sizeable literature, see e.g. the classical books (Aris, 1975; Cussler, 1976).

In particular, cross diffusion in modeling interactions among different species has attracted special attentions. Apart from the above quoted books, one can cite (Garcia-Ybarra and Calvin, 1981; Jorné, 1975; Savchik et al., 1983; Toor, 1964a,b) in physical-chemistry, (Busenberg and Travis, 1983; Kirane and Kouachi, 1993) in epidemics, (Jorné, 1977; Keller and Segel, 1970; Rosen, 1977) in ecology, population dynamics or biology and (Capasso et al., 1994) in economics.

Clonality in plant corresponds to vegetative multiplication (Bell, 1984). A clonal individual is composed of genetically identical and potentially autonomous descendants (ramets), generally linked altogether by connective stems. At first approximation, a clone can be modeled as a two dimensional tree structure more or less compact that grows from the initial ramet by branch elongations and branching. This tree structure supports resource sharing (Jónsdóttir and Watson, 1997; Stuefer et al., 2004) and storage for the clone (Suzuki and Stuefer, 1999; Suzuki and Hutchings, 1997). A connection between ramets might be broken. In that case, the clone splits: two new tree structures are formed. The main tradeoff controlling horizontal growth of plant is linked with spatial colonization and resistance to competitors. These two opposite strategies are represented by clonal plants classified as *phalanx* and *guerilla* species according to Lovett-Doust (1981). Guerilla are plants that invest resources to build distant connections in order to explore and conquest space rapidly. Their architecture is characterized by long connections between ramets which can be easily intruded by competitors. Conversely, phalanx plants are compact and produce dense clouds of ramets to extract all resources on site. Thanks to this compact structure phalanx plants resist to competition, but has limited ability to invade space.

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This ensemble of properties gives a very peculiar nature to space invasion by clonal plants where the plant architecture and the competition for space are strongly coupled.

The objective of this paper is to present a spatial competition model for clonal plant growth that combines two different mechanisms of competition. One is described by the standard underlying Kolmogorov model for two interacting populations (Brauer and Castillo-Chavez, 2000; Cushing, 1998). A second competition mechanism more specific to clonal plant growth, is expressed by adding coupling nonlinear diffusion terms, such as the motility of each species, and their capacity to exclude the other from its territory. In Kolmogorov systems, an important topic is whether every orbit starting in the interior of the first quadrant of the phase plane (i.e. the population densities are strictly positive) remains persistent. Following Czárán and Bartha (1992) and Czárán (1998), the main assumptions in the PDE model are:

(i) populations are large enough for stochastic effects not to be taken into account. In particular if one of the species has very low density, the PDE model may become irrelevant,
(ii) individuals are identical in their population dynamical attributes.

The PDE models attempt to give a phenomenological description than can reproduce spatiotemporal patterns observed in nature.

Most models of clonal plant communities are agent based models (Grimm and Railback, 2005) that take into account explicitly the plant architecture.

The question we address in this paper is the construction of a PDE model that reproduces some of the phenomenology on space competition that is typical of agent base simulations such as:

(i) no space sharing between ramets and therefore, the density of different populations has small overlaps,
(ii) space invasion has finite speed,
(iii) species with low motility have the tendency to cluster, while species with high motility disperse rather uniformly.

We will present a general PDE model for plant competition with nonlinear diffusion and examine its mathematical properties. We show in particular that, in competition systems with small spatial exclusion ability, the nonlinear diffusions do not impact the stability of the equilibrium points of the dynamical system. However, these nonlinear diffusions may modify the linear stability of these points if the considered species have strong spatial exclusion ability. We will use this model to describe the competition between two populations of clonal plants and analyze and/or compute pattern formation.

2. A general PDE model

The spatial competition model for plant growth combines two different mechanisms, one described by the standard underlying Kolmogorov model for two interacting populations (Brauer and Castillo-Chavez, 2000; Cushing, 1998)

$$\begin{cases} u'(t) = u(t)F(u(t), v(t)), \quad t > 0, \\ v'(t) = v(t)G(u(t), v(t)), \quad t > 0, \\ u(0) = u_0 > 0, \quad v(0) = v_0 > 0, \end{cases} \tag{1}$$

with another competition mechanism more adapted to clonal plant growth that expresses, by adding coupling nonlinear diffusion terms, the motility of each species, and their capacity to exclude the other from its territory.

Let us denote $(u(x,t), v(x,t))$ the time dependent spatial density of each population. The system of Partial Differential Equations (PDE) model writes:

$$\begin{cases} u_t - \mathrm{div}(\varepsilon_1 \ \mu_1(u,v)\nabla u) \ = \ u \ F(u,v), \\ v_t - \mathrm{div}(\varepsilon_2 \ \mu_2(u,v)\nabla v) \ = \ v \ G(u,v). \end{cases} \tag{2}$$

The diffusion coefficients $(\varepsilon_1, \varepsilon_2)$ are linked to the motility of each species. Let us suppose that species u would disappear in the long run in the competition mechanism with v driven by the underlying dynamical system (1). If species u has the ability to exclude strongly species v from its territory, which corresponds to $\mu_2(u,v) = 0$ if $u > 0$ and $\mu_2(0,v) > 0$, then wherever $u(x,0)$ is positive, species u will survive.

The so called community matrix associated to the Kolmogorov system

$$\begin{pmatrix} \dfrac{\partial F}{\partial u} & \dfrac{\partial F}{\partial v} \\ \dfrac{\partial G}{\partial u} & \dfrac{\partial G}{\partial v} \end{pmatrix},$$

describes, mathematically, the aggressive rate of a species versus the other in the dynamical system.

In Gurtin and MacCamy (1977), the author considers that when populations behave so as to avoid crowding it is reasonable to assume that the diffusion terms μ_1 and μ_2 are increasing functions of the population densities u and v respectively. On the other hand, since the diffusion of u is penalized where v is positive, and vice versa, it is reasonable to assume also that the diffusion terms μ_1 and μ_2 are strongly decreasing in v and u respectively. In space these generalized diffusion terms are more realistic since they allow *finite speed propagation* in space. This is in contrast to constant diffusion which leads to the standard Laplacian operator and its well known *infinite speed propagation* (Vazquez, 2006). The assumptions on the diffusions μ_1 and μ_2, on the source terms F, G and the initial data u_0, v_0 will be specified later.

Theoretically, the dependence of μ_1 in v (resp. μ_2 in u) is such that u (resp. v) cannot diffuse in spatial positions where $v > 0$ (resp. $u > 0$). Then, μ_1 and μ_2 have to behave as the following

$$\mu_1(u(x), v(x)) = \varphi_1(u(x)) \times \mathbf{1}_{\Omega\setminus\mathrm{supp}(v)}(x),$$
$$\mu_2(u(x), v(x)) = \varphi_2(v(x)) \times \mathbf{1}_{\Omega\setminus\mathrm{supp}(u)}(x),$$

for every $x \in \Omega$, where φ_1 and φ_2 are positive, increasing and smooth functions on $[0, +\infty[$. The set $\mathrm{supp}(u)$ is the closure of $\{x \in \Omega : u(x) \neq 0\}$ and for every subset ω of Ω:

$$\mathbf{1}_\omega(x) = \begin{cases} 1 & \text{if} \quad x \in \omega, \\ 0 & \text{if} \quad x \notin \omega. \end{cases} \tag{3}$$

This kind of diffusion terms leads to very complicated mathematical free boundary models which do not reflect the possible *miscibility* between the two plant populations. Then, we introduce parameters η_1 and η_2 which may reflect the ability rate of a species to exclude the other one, that is:

$$\mu_1(u,v) = \varphi_1(u)T_{\eta_1}(v), \qquad \mu_2(u,v) = \varphi_2(v)T_{\eta_2}(u), \tag{4}$$

where T_η is an approximation of the standard truncation function defined by

$$T_\eta(s) = \begin{cases} \exp\left(\dfrac{\eta^2}{s^2 - \eta^2}\right) & \text{if} \quad 0 \le s < \eta, \\ 0 & \text{if} \quad s \ge \eta. \end{cases} \tag{5}$$

This means that if the biomass u satisfies $u \ge \eta_2$, then the diffusion coefficient $\mu_2(u,v)$ vanishes and vice versa.

In the companion paper (El Hamidi et al., 2012), we have shown local and global existence results of classical solutions to System

$$\begin{cases} u_t - \operatorname{div}(\mu_1(u,v)\nabla u) = f(u,v) \text{ in } \Omega\times]0,+\infty[, \\ v_t - \operatorname{div}(\mu_2(u,v)\nabla v) = g(u,v) \text{ in } \Omega\times]0,+\infty[, \\ \nabla u\cdot\mathbf{n} = \nabla v\cdot\mathbf{n} = 0 \text{ on } \partial\Omega\times[0,+\infty[, \\ u(x,0) = u_0(x) \geq 0, \quad v(x,0) = v_0(x) \geq 0 \text{ in } \Omega, \end{cases} \tag{6}$$

- For local existence, we used the theory developed by (Amann, 1993) for semilinear reaction–diffusion systems.
- For global existence, we constructed an explicit Lyapunov functional for System (6), inspired by the work of (Kouachi, 2001).

Let us, now, present the competition systems and the absence of diffusion driven instability. Let us consider the reaction–diffusion competition system (6) in the competition case, then, we assume in the sequel the *competition conditions*:

$$\frac{\partial f}{\partial v} \leq 0 \quad \text{and} \quad \frac{\partial g}{\partial u} \leq 0. \tag{7}$$

We, also, assume the standard condition which preserves the nonnegativity of the solutions u and v

$$f(0,v) \geq 0 \quad \text{and} \quad g(u,0) \geq 0, \quad \text{for any} \quad u \geq 0, \quad v \geq 0. \tag{8}$$

In the following result, we show that *diffusion driven instability*, (or *Turing instability*), cannot occur. In particular, the presence of diffusion cannot change the linear stability of homogeneous steady state solutions of (6). The proof of that theorem is given in the annex.

Before stating this result, we introduce the temporal dynamical system associated to (6)

$$\begin{cases} u'(t) &= f(u,v), \quad t \geq 0, \\ v'(t) &= g(u,v), \quad t \geq 0. \end{cases} \tag{9}$$

Then, the homogeneous steady state solutions of (6) are the steady state solutions of (9).

Theorem 1. *Let μ_1, μ_2 be any two nonnegative and C^1 functions on $[0,+\infty[\times[0,+\infty[$. Let f and g be two C^1 functions on $[0,+\infty[\times[0,+\infty[$ satisfying the competition conditions (7) and the nonnegativity condition (8). Let (u^*,v^*) be a linearly stable steady state of (9) such that $\mu_i(u^*,v^*) > 0$. Then, (u^*,v^*) has the same linear stability nature for the system (6). That is, the presence of diffusions in (6) has no effect on the linear stability of (9). In particular, diffusion driven instability cannot hold true.*

Remark 1. Consider the system (6) with $\mu_i(u,v)$ given by (4). If $\eta_1 \gg 1$ and $\eta_2 \gg 1$, since u and v are bounded then, $\mu_i(u,v) > 0$ for every $u \geq 0$ and $v \geq 0$ for $i = 1,2$. In particular, $\mu_i(u^*,v^*) > 0$ for every equilibrium solutions. In this situation, the system behaves as in the non degenerated systems, that is, the solutions tend asymptotically to a equilibrium solution of the dynamical system.

We are going, now, to consider a special case for plant competition using the standard Lotka–Volterra–Gause competition model described in the introduction.

3. Simplified competition model for clonal plants

In this section, we will study a simple reaction–diffusion system which modelizes plant growth with spatial competition. Its aim is to give an illustration of 1 on a well known competitive system.

3.1. Lotka–Volterra-nonlinear diffusion model

We consider the following system of PDE equations:

$$\begin{cases} u_t - \operatorname{div}(\mu_1(u,v)\nabla u) = u(\alpha_1 - \beta_1 u - \gamma_1 v) \text{ in }]0,+\infty[\times\Omega \\ v_t - \operatorname{div}(\mu_2(u,v)\nabla v) = v(\alpha_2 - \beta_2 v - \gamma_2 u) \text{ in }]0,+\infty[\times\Omega \\ \nabla u\cdot\mathbf{n} = \nabla v\cdot\mathbf{n} = 0 \text{ on } [0,+\infty[\times\partial\Omega \\ u(0,x) = u_0(x), \quad v(0,x) = v_0(x) \text{ in } \Omega \end{cases} \tag{10}$$

where $\alpha_1 > 0$ denotes the growth rate of the biomass u, $\alpha_1/\beta_1 > 0$ is the carrying capacity of biomass u in the absence of v and $\gamma_2 > 0$ is the competition rate of u. The positive constants $\alpha_2 > 0$, $\beta_2 > 0$ and $\gamma_1 > 0$ are defined in the same way.

It is well known that if μ_1 and μ_2 are two positive constants then every solution of (10) converges to a equilibrium point of the associated dynamical system:

$$\begin{cases} u'(t) = u(t)(\alpha_1 - \beta_1 u(t) - \gamma_1 v(t)), \quad t > 0, \\ v'(t) = v(t)(\alpha_2 - \beta_2 v(t) - \gamma_2 u(t)), \quad t > 0, \\ u(0) = u_0 > 0, \quad v(0) = v_0 > 0, \end{cases} \tag{11}$$

System (11) has 4 equilibrium points:

$$(u_1,v_1) = (0,0)$$
$$(u_2,v_2) = \left(\frac{\alpha_1}{\beta_1}, 0\right),$$
$$(u_3,v_3) = \left(0, \frac{\alpha_2}{\beta_2}\right),$$
$$(u_4,v_4) = \left(\frac{\alpha_2\gamma_1 - \alpha_1\beta_2}{\gamma_1\gamma_2 - \beta_1\beta_2}, \frac{\alpha_1\gamma_2 - \alpha_2\beta_1}{\gamma_1\gamma_2 - \beta_1\beta_2}\right).$$

We can distinguish 4 situations:

(i) $\frac{\gamma_1}{\beta_2} < \frac{\alpha_1}{\alpha_2} < \frac{\beta_1}{\gamma_2}$
(ii) $\frac{\beta_1}{\gamma_2} < \frac{\alpha_1}{\alpha_2} < \frac{\gamma_1}{\beta_2}$
(iii) $\frac{\alpha_1}{\alpha_2} > \frac{\beta_1}{\gamma_2}$ and $\frac{\alpha_1}{\alpha_2} > \frac{\gamma_1}{\beta_2}$
(iv) $\frac{\alpha_1}{\alpha_2} < \frac{\beta_1}{\gamma_2}$ and $\frac{\alpha_1}{\alpha_2} < \frac{\gamma_1}{\beta_2}$

Notice that in all cases, the steady point $(0,0)$ is linearly unstable. Let us describle the (linear) nature of the other steady states. In case (i), (u_2,v_2) and (u_3,v_3) are saddle points and (u_4,v_4) is asymptotically stable. In case (ii), (u_2,v_2) and (u_3,v_3) are stable and (u_4,v_4) is a saddle point. In this case, the line of equation

$$v = \frac{\alpha_1\gamma_2 - \alpha_2\beta_1}{\alpha_2\gamma_1 - \alpha_1\beta_2}\, u$$

in the plane (u,v) is a separatrix, which divides the quadrant $[0,+\infty[\times[0,+\infty[$ into the basins of attraction of (u_2,v_2) and (u_3,v_3) respectively.

In case (iii), the steady point $(u_4,v_4) \notin [0,+\infty[\times[0,+\infty[$ and its nature has no effect on the steady states (u_2,v_2) and (u_3,v_3) which are, respectively, stable and saddle-point. In case (iv), the same situation occurs but the nature of (u_2,v_2) and (u_3,v_3) is reversed.

Let us conclude with the following special situation for the dynamical system

$$\begin{cases} u'(t) = \eta_1 u(t)(1 - \beta u(t) - \gamma v(t)), \quad t > 0, \\ v'(t) = \eta_2 v(t)(1 - \beta u(t) - \gamma v(t)), \quad t > 0, \\ u(0) = u_0 > 0, \quad v(0) = v_0 > 0, \end{cases} \tag{12}$$

This degenerated case, $\frac{\gamma_1}{\beta_2} = \frac{\alpha_1}{\alpha_2} = \frac{\beta_1}{\gamma_2} = \frac{\eta_1}{\eta_2}$, implies that each species has the same intra-specific and inter-specific competition performance, however, one species may grow faster at a faster time scale than the other. It follows that

$$\eta_2 \frac{u'(t)}{u(t)} = \eta_1 \frac{v'(t)}{v(t)}, \quad \text{for every } t > 0$$

and hence

$$\left[\frac{u(t)}{u_0}\right]^{\eta_2} = \left[\frac{v(t)}{v_0}\right]^{\eta_1}, \quad \text{for every } t > 0.$$

On the other hand, the system (12) admits the linear set of steady states

$$\beta u + \gamma v = 1, \qquad u \geq 0, \quad v \geq 0, \tag{13}$$

that ends with the two equilibrium points of each equation separately

$$u = \frac{1}{\beta}, \quad \text{when } v = 0,$$

or

$$v = \frac{1}{\gamma}, \quad \text{when } u = 0.$$

We observe that the initial condition (u_0, v_0) can be set such as each steady solution of this set (13) can be reached.

On the contrary, if the two species differ by their nutrient requirement in the same proportion as we have seen before, i.e.

$$\begin{cases} u'(t) = \eta_1 u(t)(\alpha_1 - \beta u(t) - \gamma v(t)), & t > 0, \\ v'(t) = \eta_2 v(t)(\alpha_2 - \beta u(t) - \gamma v(t)), & t > 0, \\ u(0) = u_0 > 0, \quad v(0) = v_0 > 0, \end{cases} \tag{14}$$

with

$$\alpha_1 \neq \alpha_2,$$

the species that gets the poorest environment support for its specific needs, gets wiped out by the other one.

We are going, now, to exercise with numerical simulations, the influence of the spatial competition with respect to nonlinear diffusions on the competition mechanism driven by the Lotka–Volterra system (11).

3.2. Numerical simulation of a benchmark problem

We consider the following benchmark case:

$$\begin{cases} u_t - \operatorname{div}(\varepsilon_1 \ \Phi(v)\nabla u) \ = \ u \ (\alpha_1 - \beta_1 u - \gamma_1 v), & (x, y) \in \Omega, \ t > 0, \\ v_t - \operatorname{div}(\varepsilon_2 \ \Phi(u)\nabla v) \ = \ v \ (\alpha_2 - \beta_2 v - \gamma_2 u), & (x, y) \in \Omega, \ t > 0, \end{cases} \tag{15}$$

with homogeneous Neumann boundary conditions on $\partial\Omega$,

$$\nabla u \cdot \mathbf{n} = \nabla v \cdot \mathbf{n} = 0 \text{ on } [0, +\infty[\times \partial\Omega.$$

The nonlinear diffusion coefficient is the smooth function

$$\Phi(s) = C \exp\left(\frac{\eta^2}{s^2 - \eta^2}\right), \quad \text{if } 0 \leq s < \eta \quad \text{and} \quad \Phi(s) = 0,$$

$$\text{if } s \geq \eta > 0.$$

We set $C = \exp(1)$ and $0 \leq \Phi(s) \leq 1$.

Let's set up the parameters of the Lotka–Volterra system such that the second species v has a competitive advantage on the first species, i.e. unique stable equilibrium point

$$\left(u = 0, v = \frac{\alpha_2}{\beta_2} > 0\right)$$

with

$$\alpha_1 = 1, \ \ \alpha_2 = 1.3, \ \ \beta_1 = 1.2, \ \ \beta_2 = 0.8, \ \ \gamma_1 = 3., \ \ \gamma_2 = 1.1.$$

We test out the influence of the competitive advantage of motility by comparing the result for $\varepsilon_1 = \varepsilon_2$ with the result for $\varepsilon_1 \gg \varepsilon_2$.

Similarly, we test the competitive advantage of space exclusion that corresponds to $\eta \ll 1$ by comparing to the situation with quasi uniform diffusion $\eta \gg 1$. Figs. 1–4 show the four possible combinations of these parameters settings.

The initial condition for the benchmark problem is a collection of well separated sharp bell function

$$\exp\left[-\frac{\operatorname{dist}((x, y), M_j)}{\nu}\right],$$

that plays the role of initial "density of plant populations", centered on a set of points M_j – see Figs. 1–4 at $T = 0$. We use similar initial densities for both species.

The numerical implementation is straightforward. We use a regular Cartesian grid with a second order finite volume approximation in space and a second order Crank–Nicolson implicit time stepping.

When we run the simulation, we observe, at early time, the diffusion process that represents plant colonization. After some time, we start to see some interaction with narrow fronts layers where both species compete. This boundary layer that separates the species, is more or less diffused depending on how small is the ν value.

In these figures, we use the following convention: the space invaded by the first species (resp. the second species) is marked by black label (resp. red label). The simulation exhibit then nice pattern formation for each species.

In the figures, area where density is above 0.5, is indicated by circles 'o', while the area with lower concentration in the interval (0.1, 0.5) is indicated by dots '.'. This helps to visualize area where the first species is strongly dominant, i.e. red circle zone, areas where the second species diffuses every where, i.e. black dot point area, and overlapping zone where competition occurs. These patterns are, somewhat, similar to the spatial distribution of ramets in a phalanx–guerilla species system emerging from the simulation of prairies with the agent base model of Mony et al. (2011). Fig. 5 gives a snap shot of one of these agent based simulations between populations of phalanx and guerilla plants: we start from an equal number of roots for each plant. We assume a small probability that clones die but asexual reproduction is implemented which let the clonal plant surviving through its youngest shoot. In agent based simulations, we let the system evolve for several thousand steps, which may represent several seasons. Fig. 5 illustrates the emergence of pattern spatial formations such as the clustering in space of phalanx plants, while guerilla plants are distributed with sparsity everywhere. The prairie is, actually, the superposition of both distribution of ramets, since there is no space sharing of the cellular automata cells at the ramet location.

The next step, in this work, would be to calibrate the PDE simplified model to best fit the agent based simulation using the assimilation technique of Le Dimet et al. (2006) for example. However, this is not an easy task from the mathematical point of view, since it requires investigations about regularity of solutions to degenerated nonlinear reaction–diffusion systems with respect to the parameters to be assimilated. These investigations will be the purpose of a future publication.

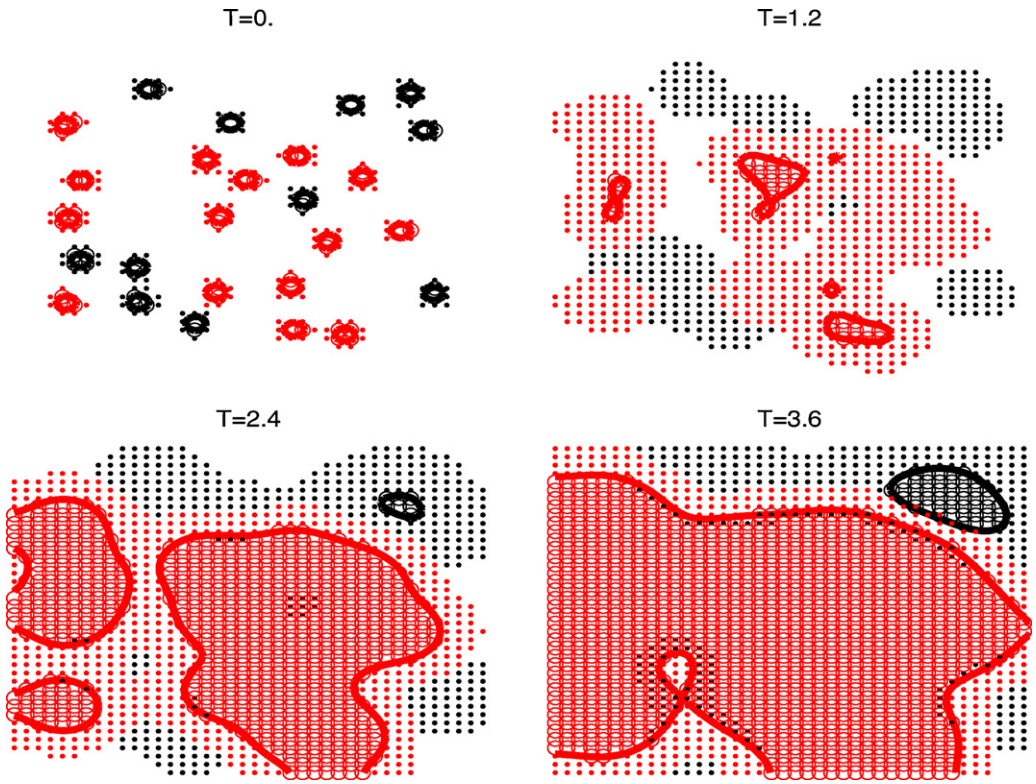


Fig. 1. Two species system with quasi uniform diffusion: $\varepsilon_1 = \varepsilon_2 = 0.001$, $\eta = 100$. First species (resp. the second species) is marked by black label (resp. red label). Density above 0.5 is indicated by circles 'o', density in the interval (0.1, 0.5) is indicated by dots'.'. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

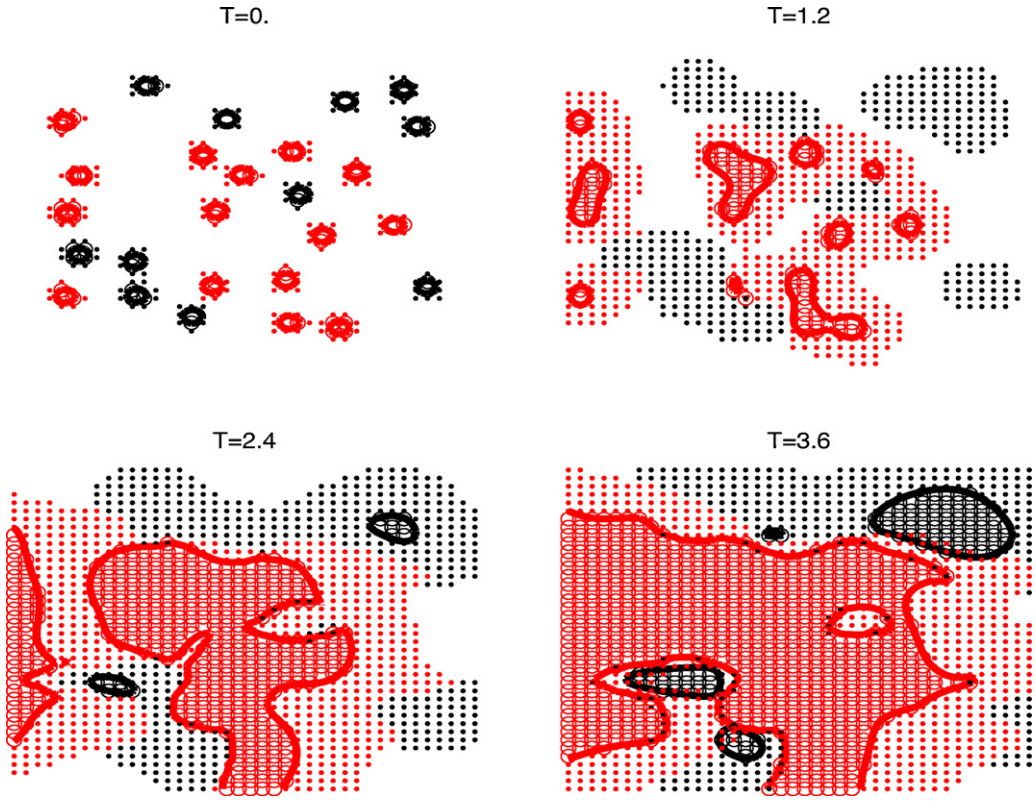


Fig. 2. Two species system with strong exclusive diffusion $\varepsilon_1 = \varepsilon_2 = 0.001$, $\eta = 0.1$. First species (resp. the second species) is marked by black label (resp. red label). Density above 0.5 is indicated by circles 'o', density in the interval (0.1, 0.5) is indicated by dots '.'. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

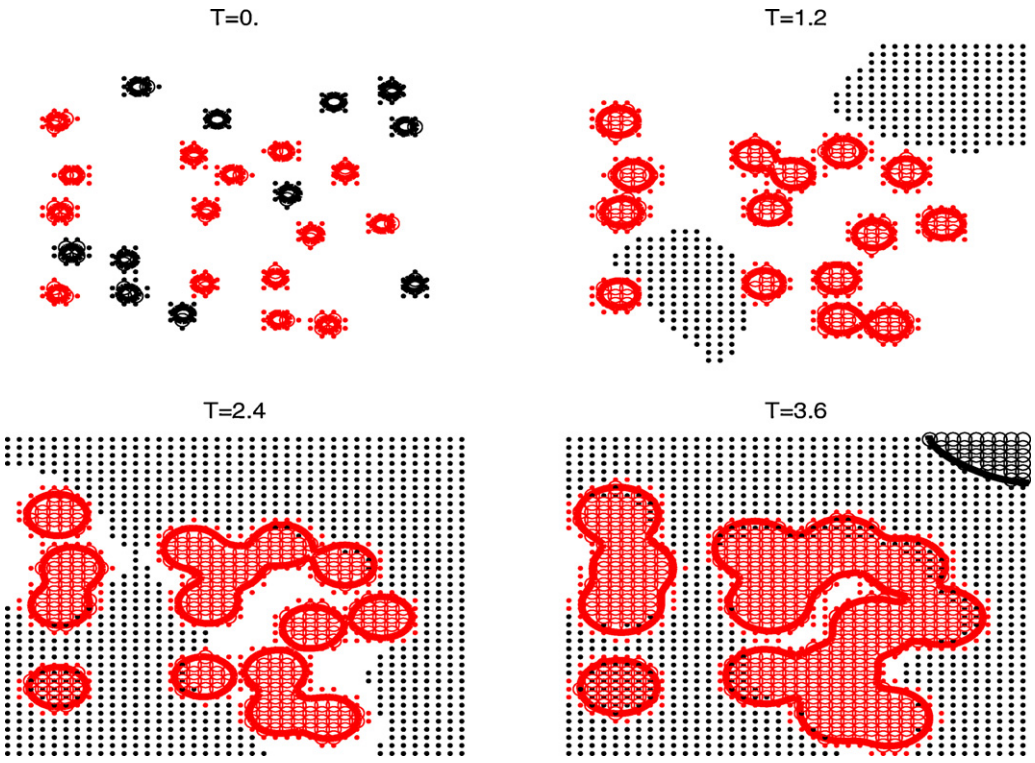


Fig. 3. First species has dominant motility $\varepsilon_1 = 0.1 \gg \varepsilon_2 = 0.0001$, $\eta = 100$, but both species have quasi uniform diffusion. First species (resp. the second species) is marked by black label (resp. red label). Density above 0.5 is indicated by circles 'o', density in the interval (0.1, 0.5) is indicated by dots '.'. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

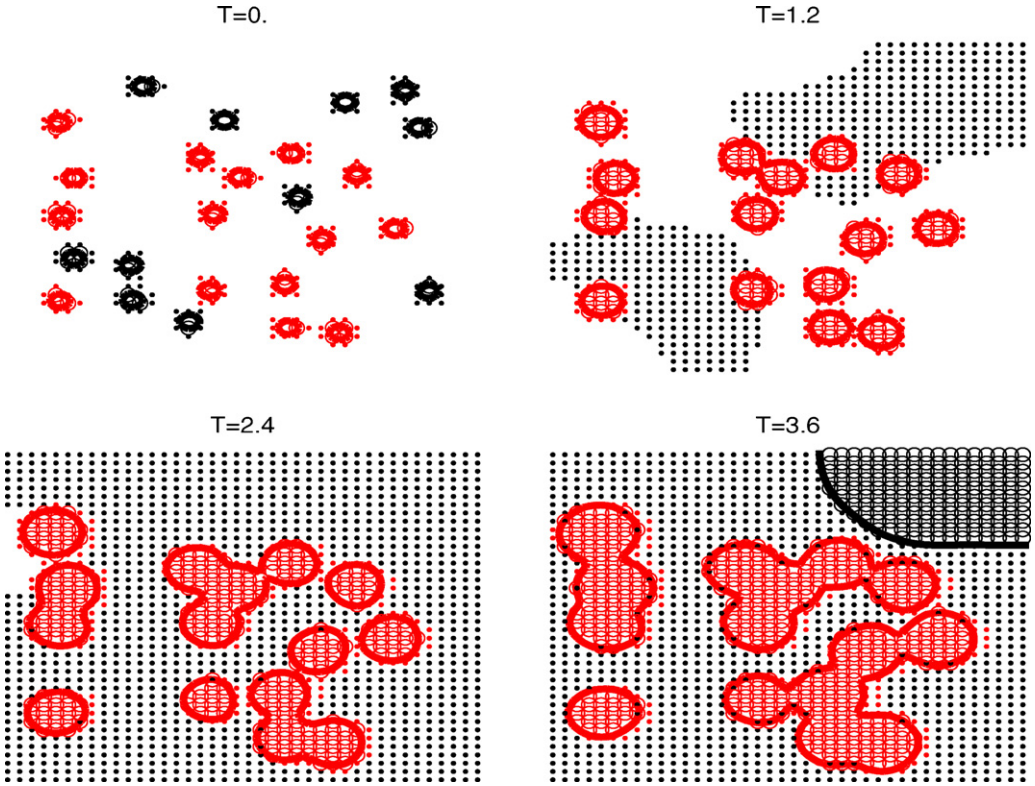


Fig. 4. First species has dominant motility $\varepsilon_1 = 0.1 \gg \varepsilon_2 = 0.0001$, $\eta = 0.1$, but both species have strong exclusive diffusion. First species (resp. the second species) is marked by black label (resp. red label). Density above 0.5 is indicated by circles 'o', density in the interval (0.1, 0.5) is indicated by dots '.'. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

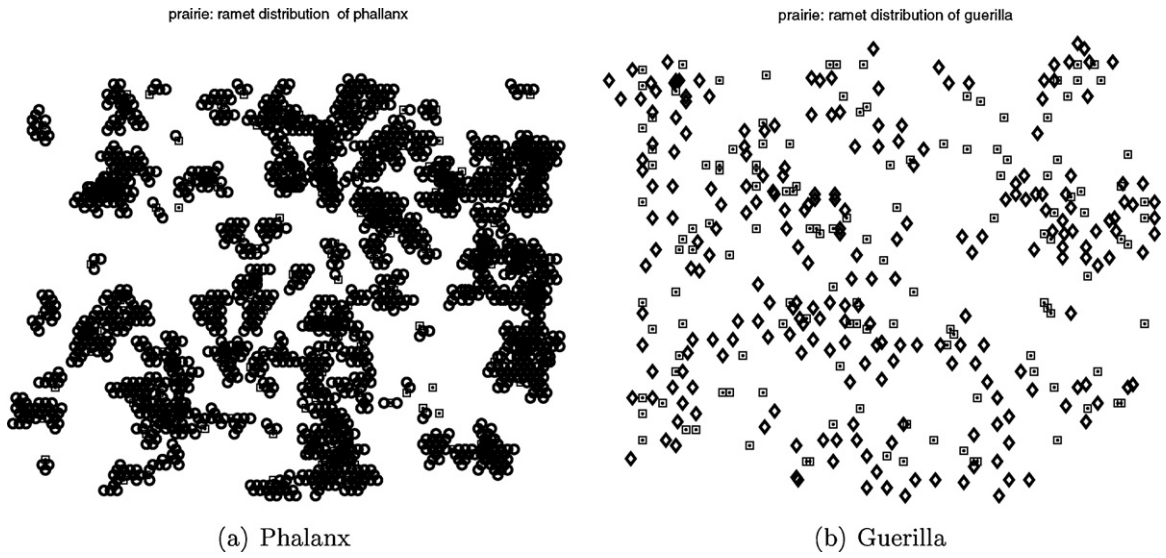


Fig. 5. Spacial distribution of ramets result with an agent base simulation of a population of guerilla and phalanx plants. (a) Phalanx, (b) Guerilla.

Let us report on the simulation result of our simplified Lotka–Volterra-nonlinear diffusion model.

As expected, with uniform spatial diffusion, i.e. $\eta \gg 1$, the most competitive species wipes out the other one that goes to extinction. Indeed, in Figs. 1 and 3, the nonlinear diffusion coefficients is positive, i.e. $\mu_i(u_k, v_k) > 0, i = 1, 2$ and $k = 1, \ldots, 4$, where (u_k, v_k) denotes the equilibrium points of (17). In particular, the unique stable equilibrium point (u_3, v_3) for (17) remains stable for (16). In Fig. 1, we say clearly the convergence of the solution to the equilibrium point (u_3, v_3).

This process might be slowed down if the less competitive species is space dominant, i.e. diffuses the fastest $(\varepsilon_1 \gg \varepsilon_2)$.

However, for strong exclusive diffusions, i.e. $\eta \ll 1$, we have $\mu_1(u_3, v_3) = 0$. In this case, 1 is not applicable and we cannot predict the stability of (u_3, v_3) from 1.

In this situation, we expect that the less competitive species can survive much better in the transient. However, as time goes to infinity, it seems that the most competitive species will, eventually, take over. As $\eta \to 0$, this process takes longer and longer time.

Eventually, for $\eta = 0$, the less competitive species will, always, survive, and perhaps gets most of the field, thanks to its initial fast space invasion. In particular, the linear stability of equilibria is completely modified by this strong degeneracy.

4. Discussion

We are going to, successively, discuss the modeling approach, its validation, and the concept of surrogate model for ecology engineering.

Density of population's modeling

The new reaction–diffusion model was designed to reproduce some of the phenomenology on space competition observed in the field and that are illustrated by agent based simulations.

The first level of competition between plants is described by a standard Kolomogorov model that can apply to many different population models.

The second level of competition is focused on spatial competitions and is more specific to clonal plant. We have introduced a new mechanism in the model by adding a nonlinear diffusion that takes into account both the motility characteristic of the plant and its capacity to exclude other species in space.

We provided a general PDE model with proof of global and local existences and analyzed the mechanism of stability and instability. It was shown mathematically that the additional competition mechanism, described by the nonlinear diffusion, cannot change the equilibrium points of the Kolomogorov model. The nonlinear source terms in the Lotka–Volterra dynamical system stays the primary mechanism of competition.

Our new model, illustrated by the benchmark problem of Section 3.2, seems to demonstrate that it can capture some of the emerging properties observed in spatial colonization with guerilla-phalanx species systems.

It should be noticed that our model can be readily generalized to clonal plant communities with multiple species:

$$W_t - \nabla \cdot (\mu(W)) = F(W),$$

where W is a vector of the density function for each species. We conjecture that the equilibrium points of the dynamical system $W_t = F(W)$, will not be affected by the nonlinear diffusion term either. Besides nonlinear diffusion, as opposed to uniform constant diffusion, will allow to obtain spatial pattern formation.

There are many ways, indeed, to improve the realism of our PDE model. For example, a digital map of the terrain Ω, may provide spatial heterogeneities that can be directly translated into the diffusion matrix. This might be also an interesting way to test the PDE model for heterogeneous patching (Oborny, 1994; Pacala and Tilman, 1994; Oborny et al., 2001).

Of course, the scale of the patching has to be larger than the scale of the clonal plant, for the PDE model to operate. This may limit, severely, the interest of the PDE model to address any interesting theoretical questions on the effect of small heterogeneities in the environment.

Another interesting question that we are currently investigating, is to add the so called "aging term" in the model. Let us consider, for example, a single species logistic model:

$$\frac{dM}{dt} = \alpha\ (\kappa - M(t))\ M(t). \tag{16}$$

Adding the effect of aging, thanks to the delay term $-\lambda\ \frac{M(t)}{M_{\max}}\ M(t - T)$,

$$\frac{dM}{dt} = \alpha\ (\kappa - M(t))\ M(t) - \lambda\ \frac{M(t)}{M_{\max}}\ M(t - T). \tag{17}$$

not only changes the dynamic of the model, but also adds interesting phenomena such as overshooting above the so called "maximum capacity" κ during the transcient.

This aging term may turn out to be essential in another aspect of the competition between clonal plants depending on if its life cycle is seasonable or not.

An interesting third kind of extension of our model would be to add, in the model, a feedback mechanism with the environment. We can take into account the complex coupling with ground flow that may, definitively, influence the supplies of nutrient, and/or add external perturbations such as grazing.

Link to agent based modeling

Most models of clonal plant communities are agent based models (Grimm and Railback, 2005) that take into account, explicitly, the plant architecture. In this bottom up modeling approach, all the basic rules on plant growth are implemented at the level of the ramet, stolon, and finally clonal plant itself. These rules describe the plant architecture growth rules and metabolic functions. Rules for intra and interspecific competition are directly expressed at the ramet level. Such model involves a large number of parameters that need to be carefully calibrated.

In addition, agent based models are rather computationally intensive applications, that have at least a linear growth of arithmetic complexity that depends on the number of ramets. Agent based computational models are essential to analyze the emerging mechanism of competition between plants. However, there are not practical, in the field as engineering application tools. One may just need to predict at the macro scale if there is coexistence between species or which plant survives. The speed of invasion of one species versus the other would be a key prediction factor.

PDEs as opposed to agent base model, do not describe the plant community at the ramet level nor require the hourly time scale. PDE models rather estimate the local density of plants and are focussing on local average (macroscopic) approximation. The PDE model may start from an underlying dynamical system, here we have chosen the Kolmogorov model, that expresses the competition, for example the competition for nutrient resources. The clonal Plant motion, thanks to growth and asexual reproduction, is taken into account by a diffusion term. In this modeling approach, the mathematical theory of traveling waves for reaction diffusion systems (Smoller, 1983) may provide a velocity of propagation of a species. These data may serve as global indicator of an ecosystem dynamic. Besides PDE models involve, in general, fewer parameters and are less computationally intensive than an agent based model. In principle, a PDE model can run on a PC in a matter of minutes and can companion ecology engineering with a simple graphic user interface showing patterns of population.

To summarize, agent based models are the perfect tool to investigate ecology theoretical questions but may not be computer efficient in engineering. Vice versa PDE models may not be good enough to elucidate how a clonal plant community functions but might be a computationally efficient tool as a surrogate model, once properly calibrated.

Our point of view would be to reconnect both approaches with standard assimilation techniques designed for PDEs (Le Dimet et al., 2006) and use the PDE model as a surrogate model for coarse engineering prediction.

We would like, however, to discuss what seems a fundamental conceptual limitation of our approach.

Let us consider the same benchmark problem of Section 3.2 and add periodic spatial disturbance as follows: At periodic time $t = k\,T$, $T > 0$, k positive integer, a part of the domain Ω is wiped out from its vegetation in a disk of random position $C(M, R)$ of center M and radius $R > 0$ inside the domain Ω:

$$U(x, y, k\ T) \equiv 0 \text{ and } V(x, y, k\ T) \equiv 0, \quad \forall (x, y) \in C(M, R).$$

For $\eta \ll 1$, and $\epsilon_1 \gg \epsilon_2$, the less competitive species is going to have much better chances to survive, thanks to these periodic perturbations, than the strongest one. We can even mathematically prove that for η_1 and ϵ_2 asymptotically small enough, the strongest species will go for extinction, while the weakest will always survive. This is an interesting situation that correlates with the theoretical conjecture that states that, in the absence of external perturbation, Guerilla plants are led to extinction by phalanx plants, while, with periodic perturbations of the environment, guerilla species get a strategic advantage to survive. While it is somehow satisfying to have a PDE model that supports this theory, we get absolutely no conceptual insight from our PDE model on why this conjecture should be true or not. It seems, in this situation, that an agent based model is at a better position to answer such a theoretical question.

Model validation

In engineering, computing (Roache, 1998) "verification" is the process of determining whether or not a code is "solving the equation right", while "validation" is the process of whether or not the code is solving "the right equation". Verification of the numerical code for our PDE model is, somehow, a much easier task than for an agent based model. The process of verification is mathematically well defined, while validation is, somehow, an ad hoc principle to the specific domain of science under consideration.

Validation of a model needs experimental data. It is a serious difficulty to get experimental data on clonal plant species with precisely two species only and an accurate control of environmental conditions. In addition this experimental data should be spatially explicit. We can speculate that systematic imaging of prairies with a pixel accuracy of the order of a centimeter, may provide, in some particular situations, a map of plant densities. This is the kind of data that one would need to validate a PDE model with spatially accurate competition. Unfortunately, we have not been able to access such data so far.

We propose alternatively to validate the PDE model against a realistic agent based simulation. This strategy has two advantages.

First, we can get as many spatial information as we want from an agent based simulation, provided that one has access to enough computing time. Second, comparing a PDE model to an agent based model, would be an elegant way to extract systematically global information on the dynamic of an ecological system that may have some inherent theoretical value. If our model is properly "fitted" to an agent based model of a phalanx–guerilla species system, we should be able to retrieve the two by two community matrix. Similarly we may found the coefficient of diffusion for each species ϵ_1 and ϵ_2, as well as the coefficient of "permeability" η that is characteristic of the size of the layer where species densities can overlap.

Perhaps, this would be a way to classify quantitatively clonal species produced out of agent based simulation.

The difficulty of fitting a nonlinear PDE model to an agent based model through numerical simulations should not be underestimated. This problem might be rigorously formulated with a variational formulation approach as in (Le Dimet et al., 2006). However, there should be, first, an agreement on what should be the fitness function, and what should be the mathematical formulation to filter out the noise from the data set.

As a matter of fact, agent based simulations are stochastic methods of Monte Carlo type. We observe, in general, a significant level of noise on agent based simulation data. We feel, however, that this is an interesting challenge that should result into new families of compact surrogate models based on PDE that can be used in the field with, perhaps, some engineering practical values.

Annex

Proof of Theorem 1 Since $\mu_i(u^*, v^*) \neq 0$, $i = 1, 2$, the linearized system of (6) at $\mathbf{s}^* \equiv (u^*, v^*)$ is given by

$$\begin{cases} \mathbf{w}_t = D(\mathbf{s}^*)\Delta\mathbf{w} + J(\mathbf{s}^*)\mathbf{w}, & (x,t) \in \Omega\times]0,+\infty[\\ \nabla\mathbf{w}\cdot\mathbf{n} = 0, & (x,t) \in \partial\Omega \times [0,+\infty[\end{cases} \tag{18}$$

where the diffusion and Jacobian matrices at $\mathbf{s}^*$ are given respectively by

$$D(\mathbf{s}^*) = \begin{pmatrix} \mu_1(\mathbf{s}^*) & 0 \\ 0 & \mu_2(\mathbf{s}^*) \end{pmatrix} \quad \text{and} \quad J(\mathbf{s}^*) = \begin{pmatrix} \dfrac{\partial f}{\partial u}(\mathbf{s}^*) & \dfrac{\partial f}{\partial v}(\mathbf{s}^*) \\ \dfrac{\partial g}{\partial u}(\mathbf{s}^*) & \dfrac{\partial g}{\partial v}(\mathbf{s}^*) \end{pmatrix}. \tag{19}$$

To solve this linear system in $\mathbf{w}$, we consider a general separable solution given by

$$\mathbf{w}(x,t) = A(t)\mathbf{p}(x),$$

where $A(t)$ is a scalar function of time t. After substitution of this separable function in (19) we deduce that

$$A(t) = A_0 e^{\lambda t},$$

for some real numbers λ and A_0; the vectorial function $\mathbf{p}$ is solution of the linear eigenvalue system

$$\begin{cases} D(\mathbf{s}^*)\Delta\mathbf{p} + J(\mathbf{s}^*)\mathbf{p} = \lambda\mathbf{p}, & \text{in } \Omega, \\ \nabla\mathbf{w}\cdot\mathbf{n} = 0, & \text{on } \partial\Omega. \end{cases}$$

Let $(\varphi_k)_{k\geq 1}$ be the complete orthonormal family in $L^2(\Omega)$ of the eigenfunctions of the Laplacian with Neumann homogeneous boundary conditions, and $(\sigma_k)_{k\geq 1}$ the corresponding eigenvalues respectively. Hence, the general solution of (19) is given by

$$\mathbf{w}(x,t) = \sum_{k\geq 1} A_0(\sigma_k) e^{\lambda(\sigma_k)t}\varphi_k(x)$$

where, for every $k \geq 1$, $A_0(\sigma_k)$ is a real number and $\lambda(\sigma_k)$ is a complex number satisfying the condition

$$\det[-\sigma_k D(\mathbf{s}^*) + J(\mathbf{s}^*) - \lambda I_2] = 0,$$

where I_2 is the identiy matrix of size two. This can be rewritten as

$$\det\begin{pmatrix} \dfrac{\partial f}{\partial u}(\mathbf{s}^*) - \mu_1(\mathbf{s}^*)\sigma_k - \lambda & \dfrac{\partial f}{\partial v}(\mathbf{s}^*) \\ \dfrac{\partial g}{\partial u}(\mathbf{s}^*) & \dfrac{\partial g}{\partial v}(\mathbf{s}^*) - \mu_2(\mathbf{s}^*)\sigma_k - \lambda \end{pmatrix} = 0,$$

Recall here that the linear stability of $\mathbf{s}^*$ implies:

$$\mathrm{tr}(J(\mathbf{s}^*)) \equiv \frac{\partial f}{\partial u}(\mathbf{s}^*) + \frac{\partial g}{\partial v}(\mathbf{s}^*) < 0, \tag{20}$$

$$\det(J(\mathbf{s}^*)) = \frac{\partial f}{\partial u}(\mathbf{s}^*)\frac{\partial g}{\partial v}(\mathbf{s}^*) - \frac{\partial f}{\partial v}(\mathbf{s}^*)\frac{\partial g}{\partial u}(\mathbf{s}^*) > 0. \tag{21}$$

To rend the stable equilibrium s^* instable via diffusions, the following necessary condition

$$\det[J(\mathbf{s}^*) - \sigma_k D(\mathbf{s}^*)] < 0$$

has to occur, for some $k \geq 1$, that is

$$\exists\, k \geq 1 \;:\; P_2(\sigma_k) < 0,$$

where $P_2(\sigma_k)$ is the quadratic polynom in σ_k given by

$$P_2(\sigma_k) :=\ \mu_1(\mathbf{s}^*)\mu_2(\mathbf{s}^*)\sigma_k^2 - \left(\mu_1(\mathbf{s}^*)\frac{\partial g}{\partial v}(\mathbf{s}^*) + \mu_2(\mathbf{s}^*)\frac{\partial f}{\partial u}(\mathbf{s}^*)\right)\sigma_k + \det(J(\mathbf{s}^*)).$$

Then the inequality on the discriminant:

$$\left(\mu_1(\mathbf{s}^*)\frac{\partial f}{\partial u}(\mathbf{s}^*) + \mu_2(\mathbf{s}^*)\frac{\partial g}{\partial v}(\mathbf{s}^*)\right)^2 - 4\mu_1(\mathbf{s}^*)\mu_2(\mathbf{s}^*)\det(J(\mathbf{s}^*)) > 0 \tag{22}$$

must hold true. In this situation, the lower root of P_2 has to be positive, since all eigenvalues σ_k, $k \geq 1$ are positive, except $\sigma_1 = 0$ (Neumann homogeneous boundary condition). Then, on must have

$$\mu_1(\mathbf{s}^*)\frac{\partial g}{\partial v}(\mathbf{s}^*) + \mu_2(\mathbf{s}^*)\frac{\partial f}{\partial u}(\mathbf{s}^*) > 0, \tag{23}$$

but this last condition is impossible. Indeed, (7) implies that

$$\frac{\partial f}{\partial v}(\mathbf{s}^*)\frac{\partial g}{\partial u}(\mathbf{s}^*) \geq 0,$$

on one hand, and (21) implies that $\frac{\partial f}{\partial y}(\mathbf{s}^*)$ and $\frac{\partial g}{\partial v}(\mathbf{s}^*)$ must have the same sign which is negative, due to (20). □

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